ARTIST TALK: GHOST WINDS TALKING

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1. INTRODUCTION

One day the wind blew across the open bathroom window as though it were playing harmonics on a giant flute. Ayers recorded the sounds, which included amazing glissandi and harmonics, a "staccato whistle" and a "Chinese opera soprano," which was especially expressive filtered and fluttered. Amplified rain clicks and a car door slam produced percussive effects. The singing, whistling and general moaning of the wind reminded her of ghost voices, and inspired her to contrast those sounds with the ghosts of real composers' voices in her composition, ***Ghost Winds Talking***.

2. THE GHOST VOICES

We analyzed speech samples from four composers: Harry Partch, Nadia Boulanger, Lou Harrison and Aaron Copland. We don't know whether anyone else can recognize their vocal timbres, but perhaps that is appropriate for ghosts.

The Csound additive synthesis model used in our previous *didgeridoo* synthesis design [1] was a good starting point for this project, but it needed some refinement. The new model morphs together a random group of vocal timbres as it slurs a pair of pitches.

2.1. The Spectral Snapshots

From the waveform of a fragment of Harry Partch saying "it's a kind of English" [2], we chose a group of three sounds, long "a," "k" and long "i." After picking a snapshot time for each sound, we examined the phase vocoder analysis graphs.

The vocal formants of the long "a" and "i" vowels are quite distinct. The "k" sound serves as a noisy bridge between the surrounding vowel sounds. The amplitude is distributed more evenly among the harmonics than for the two vowels, but the formants for the vowels bleed through a little bit.

2.2. Morphing the Sound Segment

We morphed the sounds together to make part of the combined sound, "a kind." Each harmonic is one component signal for the spectrum, with its own frequency and amplitude line segments for the group of morphed timbres. The vowels "a" and "i" sound reasonably like the timbre of Partch's original voice, but using this method with one spectral snapshot does not capture the full quality of the noise in the "k" sound. If we wish to replicate Harry Partch speaking those words, we'd be better off using the recorded samples. If we make Partch say something else, the words are not totally clear, but the result is useful enough for our musical purposes.

3. MUSICAL EXPRESSION OF THE DESIGN

Ayers used the ghost voices in her composition, *Ghost Winds Talking*. She stored the amplitude data from the graphs of the sounds in two groups of Csound function tables, one for consonants and one for vowels. The ghost sound typically begins with a sound from the consonant bank to simulate an articulation, and then morphs a string of vowel timbres. It also contains a pitch-bend inflection, either up or down. We used random cells in a spreadsheet program to choose the vocal sounds and vary their parameters. Repeated performances of the same sound will have varied spectra, making the performance more expressive. Although the speech sounds like unintelligible English, it is very suitable for the ghost voices used in the composition.

4. ACKNOWLEDGEMENTS

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5. REFERENCES

[1] Ayers, L. and Horner, A. 2007. "Didgeridoo Synthesis Using Timbre Morphing," *Australasian Computer Music Conference 2007*, Australasian Computer Music Conference. Canberra, Australia, pp 18-25.

[2] Partch, H. 1995. *Enclosure Two: Harry Partch: Historic Speech-Music Recordings from the Harry Partch Archives*. Innova 401.